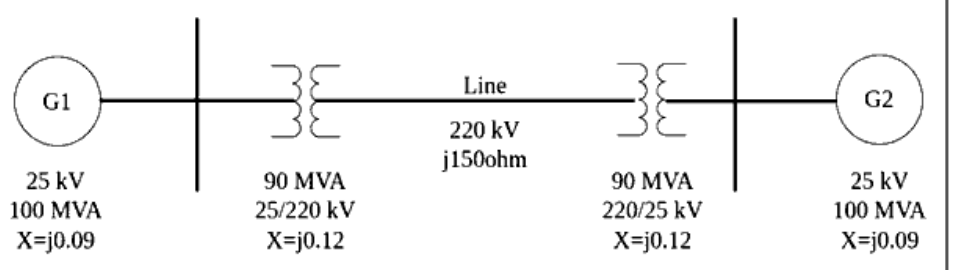
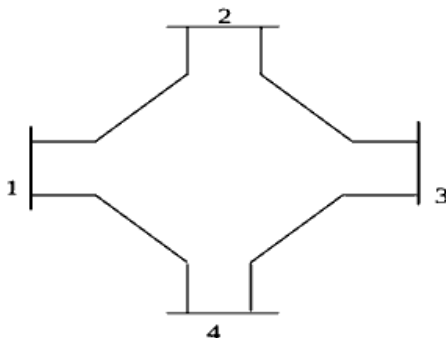


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Electrical Semester : V Subject Code & Name: BTEEC501 Power System Analysis Max Marks: 60 Date:28/01/20023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define per unit value. What are the advantages of per unit representation?	Understand	6
B)	Two loads connected in parallel are supplied from a single phase 230V source. Load 1 draws active power of 200kW at a power factor of 0.8 lagging. Load 2 draws 100kVAR at a power factor of 0.9 lag. Calculate: a. Total complex power b. Power factor of source	Application	6
C)	For the power system show in below figure. Choose 25KV as base voltage in generator G1 section and 100MVA as system base MVA. Draw the per unit reactance diagram. 	Application	6
Q.2	Solve Any Two of the following.		12
A)	Determine the bus admittance matrix for four bus system shown in figure. Neglect the shunt capacitance of line. 	Application	6

	<table><tr><th>Bus Code</th><th>Line Impedances</th></tr><tr><td>1-2</td><td>j1</td></tr><tr><td>1-4</td><td>j0.2</td></tr><tr><td>2-3</td><td>j0.8</td></tr><tr><td>3-4</td><td>j0.4</td></tr></table>	Bus Code	Line Impedances	1-2	j1	1-4	j0.2	2-3	j0.8	3-4	j0.4		
Bus Code	Line Impedances												
1-2	j1												
1-4	j0.2												
2-3	j0.8												
3-4	j0.4												
B)	Write short note on types of buses in power flow problem.	Understand	6										
C)	Explain the Gauss Seidel method of load flow analysis.	Understand	6										
Q. 3	Solve Any Two of the following.		12										
A)	Explain different types of faults in a power system. Also explain the causes and effects of faults.	Understand	6										
B)	Derive the expression for transient current on a transmission line.	Application	6										
C)	Derive the expressions of positive, negative and zero sequence voltage components of given set of unbalance voltage phasors V_a , V_b and V_c .	Application	6										
Q.4	Solve Any Two of the following.		12										
A)	The voltages of a three phase system in volts are: $V_a=400+j200$, $V_b=400-j400$, $V_c=200+j800$. Calculate the symmetrical components of the voltage.	Application	6										
B)	Define harmonics. Explain the causes and effects of harmonics in a power system.	Understand	6										
C)	What is load dispatch center? Explain the functions of load dispatch center.	Understand	6										
Q. 5	Solve Any Two of the following.		12										
A)	Draw a diagram showing interconnection of sequence network for single line to ground fault. Derive the equation for fault currents.	Application	6										
B)	An 11kV, 25MVA synchronous generator has positive, negative and zero sequence reactances of j0.12, j0.12 and j0.08 pu respectively. The generator neutral is solidly grounded. A line to ground fault occurs at the generator terminals. Determine the fault current in per unit and in Ampere. Assume that the generator was unloaded before the fault.	Application	6										
C)	Draw a diagram showing interconnection of sequence network for line to	Application	6										

	line fault. Derive the equation for fault currents.		
	*** End ***		

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